Ps 01

Erik An

<2026-03-06 Fri>

Contents

1 Problem 1 **DONE** **2**

1.1 Task 2

1.2 Solution 2

1.2.1 (1) + (2) 2

1.2.2 (3) 3

2 Problem 2 **DONE** **4**

2.1 Task 4

2.2 Solution 4

3 Problem 3 **DONE** **5**

3.1 Task 5

3.2 Solution 5

4 Problem 4 **DONE** **7**

4.1 Task 7

4.2 Solution 7

1 Problem 1 DONE

1.1 Task

For $p \in [1, \infty)$, define $d_p : \mathbb{R}^n \times \mathbb{R}^n \to [0, \infty)$ by

$$d_p(x, y) = \left(\sum_{k=1}^{n} |x_k - y_k|^p \right)^{1/p}$$

Each d_p defines a metric on $\mathbb{R}^n$.

1. Implement the function $d_p(x, y)$ in Julia.
2. Let $n = 2$. To illustrate the unit ball

   $$B_1(0) = \{x \in \mathbb{R}^2 : d_p(x, 0) \leq 1\}$$

   for $p = 1$, $p = 2$, and $p = 10$, create a grid of points in a square containing the ball and produce a scatter plot that includes only those grid points satisfying $d_p(x, 0) \leq 1$.
   Describe how the shape of the ball changes as p increases.
3. Show that

   $$d_\infty(x, y) = \max_{k=1,\ldots,n} |x_k - y_k|$$

   defines a metric on $\mathbb{R}^n$.

1.2 Solution

1.2.1 (1) + (2)

```
using Plots

d_p(p, x, y) = sum(abs.(x .- y) .^ p) ^ (1/p)

function unit_ball_scatter(p, grid_range=-1.5:0.02:1.5)
    pts = [(x,y) for x in grid_range, y in grid_range if d_p(p, [x, y], [0.0, 0.0]) <= 1]
    scatter(first.(pts), last.(pts),
            markersize=1, aspect_ratio=:equal,
            label="p=$p", title="Unit ball p=$p",
            xlabel="x_1", ylabel="x_2")
end

function main()
    plots = [unit_ball_scatter(p) for p in [1, 2, 10]]
```

```
    plot(plots..., layout=(2, 2), size=(2000, 2000))
    savefig("./assets/p_01_02.png")
end

main()
```

```
d_p (generic function with 1 method)
unit_ball_scatter (generic function with 2 methods)
main (generic function with 1 method)
"/Users/ziro2173/personal/university/01_bachelor/02_semester/01_analysis/02_exercises/set_01/ass
```

How the shape changes as p increases:

- $p = 1$: The unit ball is a diamond. The constraint $|x_1| + |x_2| \leq 1$ creates straight edges connecting the points $(\pm 1, 0), (0, \pm 1)$.
- $p = 2$: The unit ball is a circle. The Euclidian norm gives the round disk.
- $p = 10$: The unit ball is nearly square. Square with rounded corners.

1.2.2 (3)

For d_∞ to be a metric space of $\mathbb{R}^n$ it must satisfy the following properties of metric space:

1. symmetry: $\forall x, y \in \mathbb{R}^n$

$$max_{(k=1,\cdots,n)}|x_k - y_k| = max_{(k=1,\cdots,n)}|y_k - x_k|$$
$$d_\infty(x, y) = d_\infty(y, x)$$

2. triangle inequality: $\forall x, y, z \in \mathbb{R}^n$

$$max_{(k=1,\cdots,n)}|x_k - z_k| = max_{(k=1,\cdots,n)}|x_k - z_k|$$
$$max_{(k=1,\cdots,n)}|x_k - z_k| = max_{(k=1,\cdots,n)}|x_k - y_k + y_k - z_k|$$
$$max_{(k=1,\cdots,n)}|x_k - z_k| \leq max_{(k=1,\cdots,n)}|x_k - y_k| + |y_k - z_k|$$
$$max_{(k=1,\cdots,n)}|x_k - z_k| \leq max_{(k=1,\cdots,n)}|x_k - y_k| + max_{(k=1,\cdots,n)}|y_k - z_k|$$
$$d_\infty(x, z) \leq d_\infty(x, y) + d_\infty(y, z)$$

1. definiteness: $\forall x, y \in \mathbb{R}^n$

$$\begin{aligned} d_\infty(x,y) = 0 &\iff max_{(k=1,\cdots,n)}|x_k - y_k| = 0 \\ &\iff x = y \end{aligned}$$

2 Problem 2 DONE

2.1 Task

Show that for every $p \in [1, \infty)$ and for every fixed $x, y \in \mathbb{R}^n$,

$$d_\infty(x,y) \leq d_p(x,y) \leq n^{1/p}\, d_\infty(x,y)$$

,

and deduce that $d_p(x,y) \to d_\infty(x,y)$ as $p \to \infty$.

2.2 Solution

- Setup

  Let $x, y \in \mathbb{R}^n$ be fixed. Write $a_i = |x_i - y_i|$ for each i, and set $M = \max_{1 \leq i \leq n} a_i = d_\infty(x,y)$. Then

  $$d_p(x,y) = \left(\sum_{i=1}^{n} a_i^p \right)^{1/p}.$$

  Claim. For every $p \in [1, \infty)$,

  $$d_\infty(x,y) \leq d_p(x,y) \leq n^{1/p}\, d_\infty(x,y),$$

  and consequently $d_p(x,y) \to d_\infty(x,y)$ as $p \to \infty$.

- Left inequality: $d_\infty(x,y) \leq d_p(x,y)$

  Proof. Since $M = a_k$ for some index k, we have

  $$d_p(x,y) = \left(\sum_{i=1}^{n} a_i^p \right)^{1/p} \geq \left(a_k^p\right)^{1/p} = a_k = M = d_\infty(x,y),$$

  where the inequality follows from dropping non-negative terms from the sum. ■

- Right inequality: $d_p(x,y) \le n^{1/p}\, d_\infty(x,y)$

  Proof. Since $a_i \le M$ for every i, we have $a_i^p \le M^p$, and therefore

$$\sum_{i=1}^{n} a_i^p \le \sum_{i=1}^{n} M^p = n\, M^p.$$

  Taking p -th roots on both sides gives

$$d_p(x,y) = \left(\sum_{i=1}^{n} a_i^p\right)^{1/p} \le (n\, M^p)^{1/p} = n^{1/p}\, M = n^{1/p}\, d_\infty(x,y). \quad \blacksquare$$

- Deduction: $d_p(x,y) \to d_\infty(x,y)$ as $p \to \infty$

  Proof. Combining both inequalities, for every $p \in [1,\infty)$ we have

$$d_\infty(x,y) \le d_p(x,y) \le n^{1/p}\, d_\infty(x,y).$$

  Since n is a fixed positive integer,

$$n^{1/p} = e^{(\ln n)/p} \longrightarrow e^0 = 1 \quad \text{as } p \to \infty.$$

  Hence the upper bound converges to $d_\infty(x,y)$, and the Squeeze Theorem yields

$$\lim_{p\to\infty} d_p(x,y) = d_\infty(x,y). \quad \blacksquare$$

3 Problem 3 DONE

3.1 Task

Let (X, d_X) and (Y, d_Y) be metric spaces and let $f : X \to Y$ be a function. Show that f is continuous if and only if $f^{-1}(U)$ is open for all open sets $U \subseteq Y$.

(Hint: write the $\varepsilon\delta$ definition of continuity in terms of balls)

3.2 Solution

- Setup

  Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ a function. Recall that f is **continuous at** $x_0 \in X$ if

$$\forall \varepsilon > 0\ \exists \delta > 0 : d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon,$$

  or equivalently, in terms of open balls,

$$\forall \varepsilon > 0 \; \exists \delta > 0 : f(B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0)),$$

where $B_r(z)$ denotes the open ball of radius r centred at z in the appropriate metric space. We say f is **continuous** if it is continuous at every $x_0 \in X$.

Claim. f is continuous if and only if $f^{-1}(U)$ is open in X for every open set $U \subseteq Y$.

- Forward direction ($\Rightarrow$)

  Proof. Assume f is continuous. Let $U \subseteq Y$ be open and let $x_0 \in f^{-1}(U)$; we must find an open ball around x_0 contained in $f^{-1}(U)$.

  Since $x_0 \in f^{-1}(U)$, we have $f(x_0) \in U$. Because U is open in Y, there exists $\varepsilon > 0$ such that

  $$B_\varepsilon(f(x_0)) \subseteq U.$$

  Since f is continuous at x_0, for this ε there exists $\delta > 0$ such that

  $$f(B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0)).$$

  Combining both inclusions,

  $$f(B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0)) \subseteq U,$$

  which means $B_\delta(x_0) \subseteq f^{-1}(U)$. Since every point $x_0 \in f^{-1}(U)$ has such a ball, $f^{-1}(U)$ is open. ■

- Backward direction ($\Leftarrow$)

  Proof. Assume that $f^{-1}(U)$ is open in X for every open set $U \subseteq Y$. Let $x_0 \in X$ and $\varepsilon > 0$. We must find $\delta > 0$ such that $f(B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0))$.

  The open ball $B_\varepsilon(f(x_0))$ is an open set in Y, so by hypothesis

  $$f^{-1}(B_\varepsilon(f(x_0)))$$

  is open in X. Since $f(x_0) \in B_\varepsilon(f(x_0))$, we have $x_0 \in f^{-1}(B_\varepsilon(f(x_0)))$. Because this preimage is open, there exists $\delta > 0$ such that

  $$B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x_0))).$$

  Applying f to both sides,

  $$f(B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0)).$$

  This is exactly the $\varepsilon\delta$ condition at x_0. Since x_0 was arbitrary, f is continuous. ■

4 Problem 4 DONE

4.1 Task

Let (X, d_X) be a metric space and fix some $y_0 \in X$. Consider the mapping $f : X \to \mathbb{R}$,

$$f(x) = d_X(x, y_0)$$

and the balls for $\varepsilon > 0$ and $x \in X$,

$$B_\varepsilon(x) = \{y \in X : d_X(x, y) \leq \varepsilon\},$$
$$\mathring{B}_\varepsilon(x) = \{y \in X : d_X(x, y) < \varepsilon\}.$$

1. Prove that f is continuous.
2. Prove that $B_\varepsilon(x)$ is closed.
3. Prove that $\mathring{B}_\varepsilon(x)$ is open.

4.2 Solution

- Setup

  Let (X, d_X) be a metric space, fix $y_0 \in X$, and define $f : X \to \mathbb{R}$ by $f(x) = d_X(x, y_0)$. We equip $\mathbb{R}$ with the standard metric $d(a, b) = |a - b|$. For $\varepsilon > 0$ and $x \in X$ define

$$B_\varepsilon(x) = \{y \in X : d_X(x, y) \leq \varepsilon\},$$
$$\mathring{B}_\varepsilon(x) = \{y \in X : d_X(x, y) < \varepsilon\}.$$

- Part 1: f is continuous

  Proof. Let $x_0 \in X$ and $\varepsilon > 0$. We must find $\delta > 0$ such that

$$d_X(x, x_0) < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

  By the triangle inequality applied twice we obtain the **reverse triangle inequality**:

$$|d_X(x, y_0) - d_X(x_0, y_0)| \leq d_X(x, x_0).$$

  (Indeed, $d_X(x, y_0) \leq d_X(x, x_0) + d_X(x_0, y_0)$ gives one direction, and swapping the roles of x and x_0 gives the other.)

  Therefore $|f(x) - f(x_0)| \leq d_X(x, x_0)$. Choosing $\delta = \varepsilon$,

$$d_X(x, x_0) < \delta = \varepsilon \implies |f(x) - f(x_0)| \leq d_X(x, x_0) < \varepsilon.$$

  Since x_0 was arbitrary, f is continuous (in fact Lipschitz with constant 1). ■

- Part 2: $B_\varepsilon(x)$ is closed

  Proof. We show that the complement $B_\varepsilon(x)^c = \{y \in X : d_X(x,y) > \varepsilon\}$ is open.

  Let $y_0 \in B_\varepsilon(x)^c$, so $d_X(x,y_0) > \varepsilon$. Set $r = d_X(x,y_0) - \varepsilon > 0$. We claim $\mathring{B}_r(y_0) \subseteq B_\varepsilon(x)^c$.

  Let $z \in \mathring{B}_r(y_0)$, i.e.\ $d_X(z,y_0) < r$. By the triangle inequality,

  $$d_X(x,y_0) \leq d_X(x,z) + d_X(z,y_0),$$

  hence

  $$d_X(x,z) \geq d_X(x,y_0) - d_X(z,y_0) > d_X(x,y_0) - r = \varepsilon.$$

  Therefore $z \in B_\varepsilon(x)^c$. Since every point of $B_\varepsilon(x)^c$ has an open ball around it contained in $B_\varepsilon(x)^c$, the complement is open, so $B_\varepsilon(x)$ is closed. ■

  Alternative proof using Part 1. Define $g : X \to \mathbb{R}$ by $g(y) = d_X(x,y)$. By Part 1 (with x playing the role of y_0), g is continuous. Then

  $$B_\varepsilon(x) = g^{-1}((-\infty,\varepsilon]) = g^{-1}(\mathbb{R} \setminus (\varepsilon,\infty)).$$

  Since (ε,∞) is open in $\mathbb{R}$, its complement $(-\infty,\varepsilon]$ is closed, and the preimage of a closed set under a continuous function is closed. Hence $B_\varepsilon(x)$ is closed. ■

- Part 3: $\mathring{B}_\varepsilon(x)$ is open

  Proof. Let $y_0 \in \mathring{B}_\varepsilon(x)$, so $d_X(x,y_0) < \varepsilon$. Set $r = \varepsilon - d_X(x,y_0) > 0$. We claim $\mathring{B}_r(y_0) \subseteq \mathring{B}_\varepsilon(x)$.

  Let $z \in \mathring{B}_r(y_0)$, i.e.\ $d_X(z,y_0) < r$. By the triangle inequality,

  $$d_X(x,z) \leq d_X(x,y_0) + d_X(y_0,z) < d_X(x,y_0) + r = \varepsilon.$$

  Therefore $z \in \mathring{B}_\varepsilon(x)$. Since every point of $\mathring{B}_\varepsilon(x)$ has an open ball around it contained in $\mathring{B}_\varepsilon(x)$, the set is open. ■